MIL-I-25931B(ASG)
14 SEPTEMBER 1964

Superseding
MIL-I-25931A(ASG)
17 November 1961

MILITARY SPECIFICATION

INDICATOR, POSITION, LANDING GEAR, SYMBOL-INDICATING, INTEGRALLY LIGHTED

This specification has been approved by the Department of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 Scope.- This specification covers two types of hermetically sealed (see 6.3.1), integrally lighted, symbol-indicating (landing gear position) indicators.

1.2 Classification.- The indicators shall be furnished in the following Military Standard (MS) part numbers, as specified (see 6.2):

MS27202-1	Red integral lighting
MS27202-2	White integral lighting

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-L-25467	Lighting, Integral, Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Federal

FEDERAL STANDARD NO. 595	Colors

FSC 6610

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS24367	Lamp, Incandescent - Miniature Integral Lighting
MS24515	Lamp, Sub Miniature
MS25237	Lamp, Incandescent, T-1-3/4 Bulb, Midget Flange Base
MS27202	Indicator, Position Landing Gear, Symbol-Indicating, Integrally Lighted
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The indicator furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable Qualified Products List.

3.2 Data.- Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Standard parts.- Standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used, provided they conform to all requirements of this specification.

3.3.2 Commercial parts.- Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.4 Materials.-

3.4.1 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the indicator, except where magentic materials are essential.

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3.4.2 Metals.- Metals shall be corrosion resistant or shall be suitably protected to resist corrosion during normal service life.

3.4.3 Dissimilar metals.- Dissimilar metals as defined on MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing.

3.4.4 Fungus-proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent or by protective coating or hermetic sealing.

3.4.5 Corrosive fumes.- The materials, as installed in the indicator and under the service conditions specified in 3.6, shall not liberate deleterious fumes.

3.4.6 Protective treatment.- When materials are used in the construction of the indicator that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.5 Design and construction.-

3.5.1 Indicator.- The indicator shall be designed to display aircraft landing gear position by means of movable symbols. The design shall be such that, when the circuit across the actuating mechanism is completed, the proper symbol shall move without hesitation or sticking to its desired position. The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. The indicator shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases.

3.5.2 Integral lighting.- The indicator lighting shall be so designed that lamps may be accurately positioned. No one portion of the indicia shall depend solely on one lamp for its illumination. The indicator lighting system shall be built to last the normal life of the indicator and shall in no way interfere with the correct operation of the indicator. The integral lighting shall be in accordance with MIL-L-25467 for the MS27202-1 indicator, and MIL-L-27160 for the MS27202-2 indicator.

3.6 Performance.- The indicator shall conform to the tests specified under 4.6 when subjected to the following conditions:

(a) Friction: Friction error not to exceed 1/32 inch of drum movement (see 4.6.3).

(b) Position error: Position error not to exceed 1/32 inch of drum movement from the center of the aperture when tipped 90 degrees to each side and to rear from normal upright position (see 4.6.4).

(c) Insulation: A minimum resistance of 20 megohms shall exist between pins A, B, and C and the indicator case with a test voltage of 115 ±5 volts (V) at commercial frequency (see 4.6.5).

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(d) Temperature: No error shall be caused by operation at temperatures ranging from -64° to +71° C (see 4.6.14 and 4.6.15).

(e) Magnetic effect: Operating and nonoperating shall be positioned 5-1/2 inches from the center of a free magnet compass having a field intensity of 0.17 to 0.19 oersted, rotated at increments of 45 degrees from normal, without deflecting the compass more than 3 degrees (see 4.6.10).

(f) Vibration: Circular vibration in a plane inclined 45 degrees to the horizontal under the following conditions:

1. Error: Error not to exceed 1/32 inch of symbol variation or amplitude (see 6.3.2) of oscillation from the center of the aperture when operated in GEAR UP or GEAR DOWN position at an amplitude of 0.009 to 0.011 inch while the frequency is varied from 300 to 3,000 cycles per minute (cpm) (see 4.6.11).

2. Failure: No evidence of failure of operating mechanism or illuminating lamps when operated for 3 hours at an amplitude of 0.018 to 0.020 inch while the frequency is cycled between 300 and 3,000 cpm and returned to 300 cpm at a rate of 1 cycle per hour during the test (see 4.6.18).

(g) Shock: Shall withstand, with no operating failure, 18 applications of 100g impact shocks in 2 axes, each shock having a duration of 7.5 milliseconds (see 4.6.12).

(h) Indicator current: The current drawn by each operating circuit shall not exceed 0.03 ampere (amp) when connected to a 28V-dc power supply (see 4.6.13).

(i) Voltage variation: The symbol position shall not change more than 1/32 inch from the center of the aperture between operation at 20-1/2V and 30V dc (see 4.6.16).

(j) Indicator life: 20,000 cycles of operation between GEAR UP and GEAR DOWN positions at a rate of 10 to 12 cpm without damage (see 4.6.17.1).

(k) Salt spray: Withstand exposure to salt sea atmosphere for a period of 50 hours without evidence of external corrosion or deterioration.

(l) Sealing: Immersion in a de-aerated liquid in accordance with Procedure I of MIL-E-5272 without leaking, or subject to test using a helium leak detector with leak rates not exceeding those specified herein (see 4.6.7).

(m) Thermal shock: Eight cycles of exposure in water baths maintained at a temperature of 85° ±2° C and 5° ±2° C without evidence of leakage when subjected to the sealing test (see 4.6.20).

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(n) Case side pressure: A 16-pound force on each side of the case without sticking or erratic operation of the indicator (see 4.6.21).

(o) Internal examination: Examine internally for corrosion, deterioration, or damage resulting from all the tests specified herein and for compliance with the integral lighting requirements (see 4.6.22).

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.8 Case.- The case shall be in accordance with MS27202, and shall provide a hermetically sealed enclosure for the internal mechanism of the indicator. The desig of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of specia tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected. A filler tube which can be folded flush with the case or a tapered filler plug shall be used for filling purposes.

3.8.1 Case material.- The hermetically sealed case shall be constructed of a material suitable for hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.8.2 Reinforcement.- The inside of the case around the connector shall be of sufficient strength to prevent damage to the case when the connector is tightened during installation of the indicator.

3.8.3 Finish.- The case shall be finished in lusterless black conforming to color No. 37038 of Federal Standard No. 595.

3.8.4 Filling medium.- The filling medium of the heremetically sealed case shall be at least 98 percent pure, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65° C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 percent to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practicable, the 100 percent helium filling medium shall be utilized.

3.9 Lamp.- The light source for the indicator shall be two or more lamps conforming to MS25237-328, MS24367, or MS24515.

3.9.1 Lamp position.- The lamps shall be located within the hermetically sealed enclosure.

3.9.2 Lamp replacement.- Lamp replacement shall be accomplished at overhaul depots.

3.10 Cover glass.- The cover glass shall be clear and free from flaws which interfere with normal reading of the indicator. The thickness of the glass shall be entirely suitable for the purpose and shall not exceed 0.047 inch. The distance between the inside surface of the cover glass and the arc of the drum shall not exceed 1/32 inch.

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3.11 Indicating symbols.-

3.11.1 Markings.- The symbol markings shall conform to figure 1 and shall be durable to withstand usage in service. The style and proportions of numerals and letters placed on the symbols or drum shall conform to MS33558. The dial markings shall be lusterless white conforming to color No. 37875 of Federal Standard No. 595 and there shall be no color change as a result of subjecting the indicator to the tests specified herein. The dimensions of the markings shall be as specified in Table I.

TABLE I. Marking dimensions

Markings	Height or length (inch ±1/64)	Width of line or graduation (inch ±1/64)
Lettering "UP"	3/16	1/32
Unshaded part of "GEAR UNLOCKED" symbol	---	1/16
Wheel support on "GEAR DOWN"	---	3/64

3.11.2 Symbol background.- The background of the symbols shall be black conforming to color No. 37038 of Federal Standard No. 595.

3.12 Electrical power.- The indicator shall be designed for operation using 28V dc. The current required for each operating circuit shall not exceed 0.03 amp.

3.13 Electrical receptacle.- The electrical receptacle shall be Cannon part number MC-14H-10-6PN, or equal.

3.14 Electrical wiring.- The wheel position shall be wired to the electrical receptacle so that when the positive terminal of a 28V-dc source is connected to pin B and the negative terminal connected to pin A or C, the GEAR UP symbol or GEAR DOWN symbol, respectively, shall be displayed. The integral lighting shall be so wired that when 5V is applied to pin D(pin E is grounded) the lights will be energized.

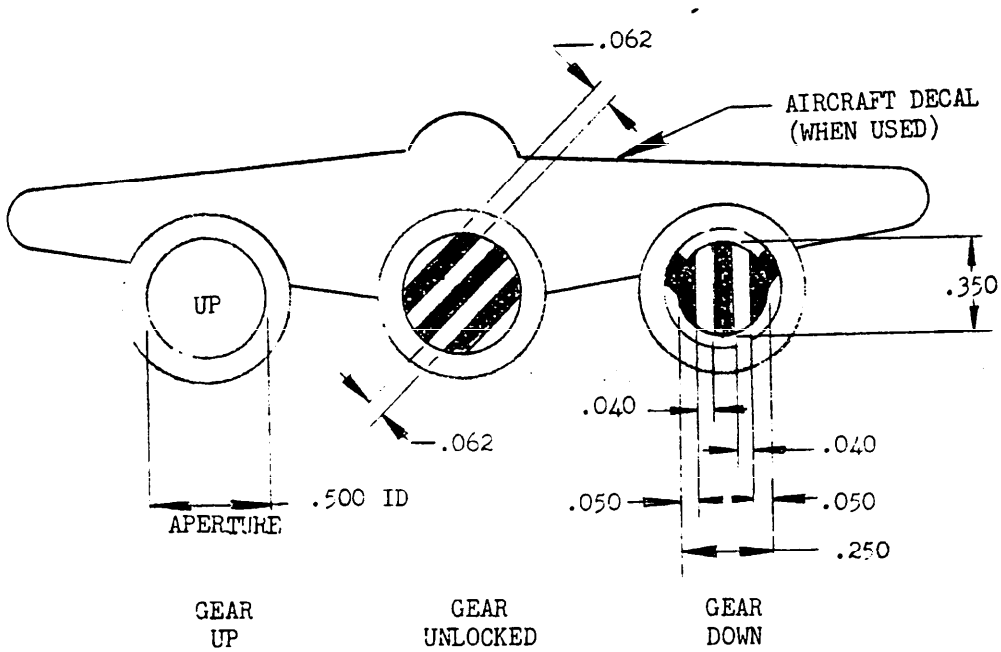
3.15 Weight.- The weight of the indicator completely assembled, less mounting screws, shall not exceed 3 ounces.

3.16 Screw threads.- Unless otherwise specified in the contract or order, the threads of all machine screws 0.060 inch or larger in diameter shall conform to MIL-S-7742.

3.17 Dial visibility.- The indicator shall in no way restrict the visibility of any graduations, numerals, symbols, or other specific markings when viewed from a cone angle of 30 degrees minimum at a distance of 30 inches from the center of the dial.

3.18 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.005

FIGURE 1. Indicating symbol

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3.18.1 Case marking.- The top side of the case shall be marked THIS END UP to facilitate proper mounting of the indicator. Decalcomanias or metal-cals shall not be used for this purpose.

3.19 Installation.-

3.19.1 Mounting screws.- The contractor shall furnish the required mounting screws and lockwashers to install the indicator. Screws shall be No. 6-32NC-2 roundhead, brass machine screws with a durable dull black finish and of proper length to permit mounting on a panel 1/8 inch thick.

3.19.2 Envelope.- An envelope containing the mounting screws and lockwashers shall be furnished with each indicator. The following information shall be printed on the face of the envelope:

IMPORTANT
THIS ENVELOPE CONTAINS
MOUNTING SCREWS AND WASHERS

3.20 Dimensions.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the indicator, they shall be held or limited accordingly.

3.21 Workmanship.- Soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, and machine screw assemblies of the indicator, including all parts and accessories, shall be neat and thorough. Assembly screws and bolts shall be tight. (The word "tight" means that the screw or bolt cannot be appreciably tightened further without damage to the screw, bolt, or threads.) Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed. Parts shall be free from burrs and sharp edges.

3.21.1 Cleaning.- The indicator shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges, as well as resin flash that may crumble, shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The inspection and testing of the indicator shall be classified:

(a) Qualification tests (4.3)
(b) Quality conformance tests (4.4)

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4.3 Qualification tests.-

4.3.1 Sampling instructions.- The qualification test samples shall consist of three indicators. Samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity (see 6.4).

4.3.2 Requalification.- If the product which has passed the qualification tests of this specification is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.3.3 Tests.- Qualification tests shall consist of all the tests specified under 4.6.

4.4 Quality conformance tests.- Quality conformance tests shall consist of individual tests and sampling plan A tests.

4.4.1 Individual tests.- Each indicator shall be subjected to the following tests conducted in the order listed:

(a)	Examination of product	(4.6.1)
(b)	Room temperature operation	(4.6.2)
(c)	Friction	(4.6.3)
(d)	Position error	(4.6.4)
(e)	Insulation	(4.6.5)
(f)	Lighting operation	(4.6.6)

4.4.2 Sampling tests.-

4.4.2.1 Sampling plan A.- One indicator selected at random from each 100 or less produced on the contract or order shall be subjected to the following tests conducted in the order listed:

(a)	Individual tests	(4.4.1)
(b)	Low temperature operation	(4.6.8)
(c)	High temperature operation	(4.6.9)
(d)	Magnetic effect	(4.6.10)
(e)	Vibration error	(4.6.11)
(f)	Shock	(4.6.12)
(g)	Indicator current	(4.6.13)
(h)	Integral lighting	(4.6.17.2)

4.4.2.2 Sampling plan B.- Unless otherwise specified in the contract or order (see 6.2), 3 indicators selected at random from the first 10 items produced on the contract or order shall be subjected to the following tests conducted in the order listed:

(a)	Sampling plan A	(4.4.2.1)
(b)	Low temperature exposure	(4.6.14)
(c)	High temperature exposure	(4.6.15)
(d)	Voltage variation	(4.6.16)
(e)	Indicator life	(4.6.17.1)
(f)	Vibration failure	(4.6.18)
(g)	Salt spray	(4.6.19)
(h)	Sealing	(4.6.7)
(i)	Thermal shock	(4.6.20)

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(j)	Case side pressure	(4.6.21)
(k)	Sampling plan B tests of MIL-L-25467 or MIL-L-27160	
(l)	Internal examination	(4.6.22)

4.4.2.3 Rejection.- When one item selected from a production run fails to conform to the requirements of this specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.4.2.3.1 Individual tests may continue.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of items on hand or later produced shall not be made until it is determined that items conform to this specification.

4.4.3 Defects in items already accepted.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them. No items or test reports shall be submitted to the procuring activity for approval until all tests have been accomplished and all items tested conform to specification requirements.

4.5 Test conditions.-

4.5.1 Standard atmospheric conditions.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified conditions.

4.5.2 Vibration.- Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 5 and 50 cps and shall subject the indicator to vibration so that a point on the indicator case will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is equal to the amplitude specified herein.

4.5.3 Attitude.- Unless otherwise specified in the contract or order, the indicator shall be tested in its normal operating position (front face of indicator in vertical plane).

4.6 Test methods.-

4.6.1 Examination of product.- Each indicator shall be inspected to determine compliance with this specification with respect to material, workmanship, and envelope.

4.6.1.1 Dial visibility.- The indicator shall be checked to assure conformance to 3.17.

4.6.2 Room temperature operation.- The indicator shall be connected to a 28V dc source and to a single-pole, double-throw switch with an OFF position. Pin B shall be connected to the positive of the 28V source. Pins A and C shall be connected to opposite sides of the double-throw switch. The arm of the switch shall be connected to the negative (ground) of the 28V source so that when pin A is connected to the ground through the switch, the GEAR UP symbol will be displayed and when pin C is

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connected to ground, through the switch , the GEAR DOWN symbol will be displayed. When the switch is in the OFF position (neither pin A or C grounded), the gear unlocked (barber pole) symbol will be displayed. The room temperature operation test shall consist of cycling the indicator, using the double-throw switch in the following sequence: OFF, pin A to ground, OFF, pin C to ground, OFF. The sequence shall be repeated for a total of three cycles. At each switch position, the indicator shall be observed. The correct symbol shall appear for each switch position within 1 second after the switch is thrown. The center of the symbol shall be within 1/3 inch of the center of the aperture.

4.6.2.1 Conflicting signals.- Upon completion of the test specified in 4.6.2, the switch shall be replaced with two single-pole switches. The indicator shall be connected to the 28V dc source, with the B terminal positive and common, and the A and C terminals each made negative through a switch. The indicator shall be cycled in accordance with the following sequence:

(a) Close switch A (making terminal A negative). The GEAR UP indication shall appear.
(b) Close switch C, with switch A still closed. The GEAR UP indication shall remain.
(c) Open switch A. GEAR DOWN shall appear.
(d) Again close switch A. The GEAR DOWN indication shall remain.
(e) Open switch C. The indication shall change from GEAR DOWN to GEAR UP position.

4.6.3 Friction.- The indicator shall be operated as above, except the indication before and after tapping shall be noted. The indicator shall be in the normal upright mounted position and shall be tapped only after the drum has come to rest (see 3.6(a)).

4.6.3.1 The difference in indication before and after tapping shall not exceed 1/32 inch of drum movement.

4.6.4 Position error.- The indication shall be noted with the indicator energized in the GEAR UP, GEAR UNLOCKED, and GEAR DOWN indication for the following positions (see 3.6(b)):

(a) In normal upright position with the dial vertical.
(b) Tipped 90 degrees to the rear with dial horizontal and face up.
(c) Rotated 90 degrees to the right on its horizontal axis with the dial in a vertical plane.
(d) Rotated 90 degrees to the left on its horizontal axis with the dial in a vertical plane.

The maximum position deviation from the normal position shall not exceed 1/32 inch of drum movement.

4.6.5 Insulation.- 115 ±5V shall be applied between pins A, B, and C of the electrical receptacles and the metal parts of the indicator case. A minimum resistance of 20 megohms shall exist (see 3.6(c)).

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4.6.6 Lighting operation.- The lighting system shall be excited with 5V and examined to ascertain that all lamps function properly.

4.6.7 Sealing.- The indicator shall be subjected to immersion tests in accordance with Procedure I of MIL-E-5272. If a mass spectrometer helium leak detector is utilized, the following leak rates shall apply (see 3.6(l)).

(a) Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.01 micron cubic foot per hour at a pressure differential of 1 atmosphere.

(b) Where 100 percent helium is used as the filling medium, the leak rate shall not exceed 0.1 micron cubic foot. per hour.

4.6.8 Low temperature operation.- The indicator shall be subjected to a temperature of -54° ±2° C for 4 hours. At the end of this period, and while still at this temperature, the indicator shall be tested as specified in 4.6.2. The correct symbol shall appear within 3 seconds after the switch is positioned, and its center shall be within 1/32 inch of the center of the aperture. The indicator shall be returned to room temperature and be subjected to and pass the tests specified in 4.6.2 and 4.6.3.

4.6.9 High temperature operation.- The indicator shall be subjected to a temperature of 71° ±2° C for 4 hours. At the end of this period, and while still at this temperature, the indicator shall be tested as specified in 4.6.2. The correct symbol shall appear within 1 second after the switch is positioned, and its center shall be within 1/32 inch of the center of the aperture. The indicator shall be returned to room temperature and be subjected to and pass the tests specified in 4.6.2 and 4.6.3.

4.6.10 Magnetic effect.- The indicator, first, not operating, and second, operating on rated voltage (with lamps illuminated) shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center compass. Starting directly under the compass, the indicator shall be held in positions zero degree, 45 degrees, 90 degrees, 135 degrees, 180 degrees, 225 degrees, 270 degrees, and 315 degrees from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in its normal position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3 degrees (see 3.6(e)).

4.6.11 Vibration error.- The indicator shall be mounted on the vibration stand in its normal operating position. The indicator shall be electrically connected for operational and lighted conditions, and tested in the GEAR UP and GEAR DOWN positions to vibration at an amplitude setting between 0.009 and 0.011 inch and at frequencies varying from 300 to 3,000 cpm. During vibration, the maximum amplitude of oscillation and variation of the movable symbol shall not exceed 1/32 inch. The lamps shall be operating at the completion of this test (see 3.6(f) 1.).

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4.6.12 Shock.- The indicator shall be tested for shock in accordance with Procedure IV of MIL-E-5272. The test specimen shall be subject to 18 impact shocks of 100g, each shock impulse having a time duration of 7.5 ±1 milliseconds (see 3.6(g) The intensity shall be within ±10 percent when measured with a filter having a band width of 5 to 100 cps. The maximum gravity shall be reached in approximately 3.75 milliseconds. The shock shall be applied in the following directions:

(a) Vertical axis, three shocks in each direction.
(b) Horizontal axis (major and minor), three shocks in each direction.

This test shall be accomplished on three indicators, on one indicator with the power off, on one with pins A and B energized, and on one with pins C and B energized. Upon completion of this test, the indicators shall be tested and shall comply with all the tests specified in 4.4.1.

4.6.13 Indicator current.- The indicator shall be tested, using 28V-dc power to determine that the current required for each operating circuit shall not exceed 0.03 ampere (see 3.6(h)).

4.6.14 Low temperature exposure.- The indicator shall be subjected to a temperature of -64° ±2° C for a period of 48 hours. While still at this temperature, the indicator shall be operated throughout its range without exhibiting sticking or erratic characteristics. The temperature shall then be raised to -54° ±2° C for a period of 24 hours. The indicator shall then be subjected to the test specified in 4.6.8 while at the -54° C temperature. There shall be no damage which would adversely affect subsequent operation.

4.6.15 High temperature exposure.- The indicator shall be subjected to a temperature of 71° ±2° C for 24 hours. At the end of this period and while at this temperature, the indicator shall be subjected to and meet the requirements specified in 4.6.2. The indicator shall be returned to room temperature and shall be subjected to and meet the requirements specified in 4.6.2 and 4.6.3. There shall be no damage which would adversely affect subsequent operation.

4.6.16 Voltage variation.- The indicator shall be operated at a reduced voltage of 22-1/2V and an increasing voltage of 30V. The indication shall not change more than 1/32 inch of drum movement between operation at the reduced voltage and operation at the increased voltage. The indicator shall be tapped lightly after each change in voltage (see 3.6(i)).

4.6.17 Life.-

4.6.17.1 Indicator.- The indicator shall be operated for 20,000 cycles at a rate of 10 to 12 cpm. A cycle shall consist of movement from the GEAR UNLOCKED position, to the GEAR UP position, to the GEAR UNLOCKED position, to the GEAR DOWN position, and back to the GEAR UNLOCKED position (see 3.6(j)). After this test, the indicator shall be subjected to and shall pass the tests specified in 4.6.2 and 4.6.3.

4.6.17.2 Integral lighting.- The integral lighting system shall be tested in accordance with MIL-L-27160 or MIL-L-25467, as applicable.

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4.6.18 Vibration failure.- The indicator shall be subjected to the vibration test Procedure IV of MIL-E-5272 for 3 hours. The test shall be conducted while the indicator is being operated on 28V dc and while the bulbs are illuminated with 5V. The indicator shall be subjected to vibration with an amplitude between 0.018 and 0.020 inch. After the vibration, the indicator shall be subjected to the tests specified in 4.6.2 and 4.6.3. In addition, no looseness in the mechanism nor damage to any part of the indicator shall result from the vibration. The internal lamps shall remain illuminated during and after the test (see 3.6(f)2.).

4.6.19 Salt spray.- The indicator shall be subjected to the salt spray test Procedure I of MIL-E-5272 for 50 hours. Following this test, the indicator shall be examined and there shall be no evidence of external corrosion or deterioration. The indicator shall then be subjected to the tests specified in 4.4.1, except (f). The external surface of the indicator may be washed with distilled water and air dried prior to the insulation test.

4.6.20 Thermal shock.- The indicator shall be subjected to eight cycles of exposure to water at 85° ±2° C and 5° ±2° C without evidence of moisture penetration or damage to coating or enclosure (see 3.6(m)). Each cycle of the test shall consist of immersing the indicator in water at 85° C for 30 minutes and then, within 5 seconds of removal from the bath, immersing for 30 minutes in the other water bath maintained at 5° C. This cycle shall be repeated continuously, one cycle following the other, until eight cycles have been completed. Following this test, the indicator shall be subjected to the test specified in 4.6.7. No indicator leakage shall occur as a result of this test.

4.6.21 Case side pressure.- A 16-pound force shall be applied to the approximate geometric center of each side of the case. The force shall be applied through the flat end of a 1/4, ±1/32-inch diameter rod. The application of the force shall not hinder free movement of the symbol (see 3.6(n)) nor affect the operation or the illumination from the lamps.

4.6.22 Internal examination.- The indicator, following all tests specified herein, shall be opened and examined internally. There shall be no evidence of deterioration or damage to the mechanism or lighting functions (see 3.6(o)).

4.7 Preservation, packaging, packing, and marking.- Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- The indicator shall be packaged in accordance with MIL-P-7936. Preservation shall be in accordance with Method III of MIL-P-116.

5.1.2 Level C.- The indicator shall be preserved and packaged in accordance with MIL-P-7936.

5.2 Packing.-

5.2.1 Levels A, B, and C.- The indicator shall be packed in accordance with levels A, B, and C as applicable, of MIL-P-7936.

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5.3 Marking.- Interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The individual landing gear position indicator is intended for use on aircraft with retractable landing gear where indication of the position of the landing gear is desired and where an integrally lighted indicator is required. The individual indicators may be mounted and grouped as desired on a decalcomania outline of the aircraft on the instrument panel.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) MS part No. of the indicator desired (see 1.2).
(c) Data requirements (see 3.2).
(d) Whether sampling plan B is to be omitted (see 4.4.2.2).
(e) Levels of packaging and packing desired (see 5.1 and 5.2).

6.3 Definitions.-

6.3.1 Hermetic seal.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.3.2 Amplitude.- Whenever the word "amplitude" is specified it means the extent of motion as measured from one extreme to the opposite extreme.

6.4 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group (SEG), Wright-Patterson Air Force Base, Ohio 45433,and information pertaining to qualification of products may be obtained from that activity.

6.5 International standardization.- Certain provisions (see 3.11.1) of this specification are the subject of international standardization agreement ABC 10/A (STANAG 3329). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

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Custodians:
  Navy - WP
  Air Force - (11)
International interest (see 6.5)

Preparing activity:
  Air Force - (11)

Reviewer activity:
  Navy - WP
  Air Force - (11)

User activity:
  Navy -
  Air Force -

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DoD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-I-25931B(ASG) Indicator, Position, Landing Gear, etc.

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity)	DATE

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Wright-Patterson AFB, Ohio 45433

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